



# Università di Catania

## Big Data Analytics Assignment

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## Introduction

This project has been developed following a **bottom-up** approach. Such an approach involves the initial construction of **a single data mart**, consistent with the illustrative nature of the project, which does not require the complexity involved in developing and integrating multiple data marts into a complete data warehouse system.

Specifically, the goal was to build an infrastructure capable of supporting the analysis of financial transactions related to publicly traded companies, through the use of **OLAP techniques** and an **interactive dashboard**.

## Dataset Description

For this project, the following three CSV files are used:

- `account-statement-1-1-2024-12-31-2024.csv`

This file contains financial transactions. Each row is identified by an `IDTransaction` and is associated with a `Date`, a `TransactionType`, a `Symbol` (referring to the financial security), and a `Unit` value indicating the quantity traded. *The data is granular at the single transaction level.*

| Feature                      | Description                                    |
|------------------------------|------------------------------------------------|
| <code>IDTransaction</code>   | Unique identifier of the transaction           |
| <code>Date</code>            | Date and time of the transaction               |
| <code>TransactionType</code> | Type of transaction (BUY, SELL, DIVIDEND)      |
| <code>Symbol</code>          | Company symbol associated with the transaction |
| <code>Unit</code>            | Number of units traded                         |

- `symbols.csv`

This file contains information about financial securities and forms the basis for building the symbol dimension. Each row describes a company through its symbol, name, economic sector, industry, and country of origin.

| Feature                   | Description                           |
|---------------------------|---------------------------------------|
| <code>symbol</code>       | Unique identifier of company          |
| <code>company_name</code> | Name of the listed company            |
| <code>sector</code>       | Economic sector of the company        |
| <code>industry</code>     | Industry to which the company belongs |
| <code>country</code>      | Country of origin (ISO name)          |

- `country.csv`

This file provides standardized geographic information useful for regional analysis of transactions. Columns include ISO codes (both alphanumeric and numeric), geographic hierarchy levels (region, sub-region, intermediate region), and corresponding numeric codes.

| Feature                  | Description                             |
|--------------------------|-----------------------------------------|
| name                     | Official name of the country            |
| alpha-2                  | ISO 3166-1 alpha-2 code                 |
| alpha-3                  | ISO 3166-1 alpha-3 code                 |
| country-code             | ISO numeric country code                |
| iso_3166-2               | Administrative subdivision code         |
| region                   | Macro geographic region (e.g., Asia)    |
| sub-region               | Sub-region (e.g., Southern Asia)        |
| intermediate-region      | Intermediate region (if present)        |
| region-code              | Numeric code of the region              |
| sub-region-code          | Numeric code of the sub-region          |
| intermediate-region-code | Numeric code of the intermediate region |

## Setting goal

The main goal of this project is twofold:

- **OLAP Queries**, to answer specific analytical questions:
  - What are the top 5 sectors by number of SELL transactions in China during 2024?
  - What are the top 5 industries by number of BUY transactions in Q4 of 2024?
  - Rank all quarters of 2024 by total number of transactions (BUY + SELL).
- **Interactive dashboard**, developed with **Streamlit**, enabling exploratory navigation of the data. The expected features include:
  - A date range filter, with default values set to:
    - \* Start date: 01/01/2024
    - \* End date: 31/12/2024
  - A line chart showing the total number of transactions (BUY + SELL) over time.
  - Three bar charts showing (for the same date range and filtered data):
    - \* The top 3 traded symbols by transaction count.
    - \* The top 5 sectors by transaction count.
    - \* The top 5 industries by transaction count.

The implementation of this system requires careful planning of the architecture, data model, and ETL processes to ensure efficient analytical performance and data quality.

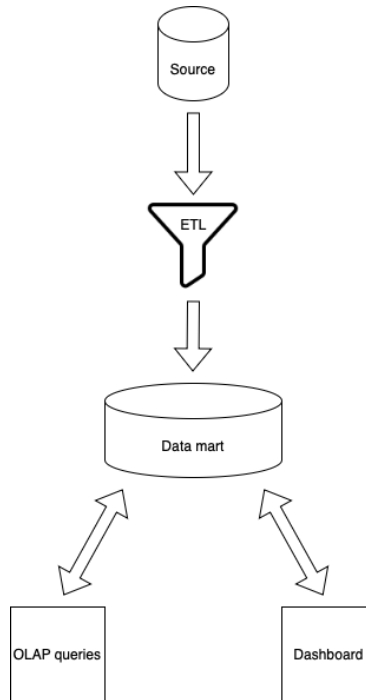


Figure 1: Architecture

## Designing Infrastructures

A simplified architecture was adopted for this project, consisting of the following logical layers:

- **Source layer:** composed of three CSV files containing raw data on financial transactions, company symbols, and geographic information.
- **ETL process:** includes the extraction, transformation, and loading of data. This process was implemented in a simple and direct way, without using a persistent staging layer.
- **Data warehouse layer:** consists of a single thematic data mart, modeled using a star schema, which serves as the foundation for OLAP analysis and dashboard generation.
- **Analysis layer:** includes analytical queries and an interactive dashboard developed with Streamlit.

No intermediate *Data Staging Layer* or *Reconciled Data Layer* was implemented, as the simplicity and clarity of the provided datasets—together with the illustrative nature of the project—made it unnecessary to introduce additional transformation or persistent integration layers.

Although minimal, this architecture is suitable for the objectives of the project, which aims to demonstrate the fundamental principles of analytical modeling and the construction of a business intelligence pipeline for educational purposes. In real-world enterprise scenarios, a more structured three-layer solution (including the reconciled layer) would be preferred to ensure scalability, governance, and data reusability.

## Designing and Developing Data Marts

The project was developed following a *requirement-driven* approach. In this methodology, the design of the data mart is guided by the informational needs expressed by end users, and the necessary data structures are subsequently identified to meet those needs.

In this specific case, the analytical queries specified in the assignment (e.g., “Top 5 sectors for SELL transactions in China in 2024”) guided the following aspects:

- the selection of relevant informational entities;
- the definition of measures and dimensions;
- the design of the multidimensional model.

The design of the data mart followed these main phases:

1. **Requirement Analysis**
2. **Conceptual Design**
3. **Logical Design**
4. **Implementation using Python**

This choice is justified by the illustrative and simplified nature of the project, which aims to build a system for educational purposes. As such, it is not necessary to carry out all the steps of the full methodology (such as physical design, workload and data volume, or the creation of a data staging layer).

In an industrial or production environment, however, these additional phases would be essential to ensure the scalability, robustness, and maintainability of the entire business intelligence system.

## 1 Requirement Analysis

The requirement analysis phase, consistent with the *requirement-driven* approach, involves defining the information necessary to meet the analytical goals of the system. In the literature, there are advanced methodologies, such as *goal-oriented* models (e.g., *Tropos*), that enable a structured formalization of users’ informational goals and their translation into multidimensional concepts.

In this project, however, a more direct and pragmatic strategy was adopted, based on the explicit analysis of the requests outlined in the assignment (OLAP queries and dashboard functionalities), in order to maintain focus on the essential aspects of designing and building the data mart in a simplified educational context.

## Identifying Relevant Information

The requirement analysis was conducted starting from the OLAP queries and the expected functionalities of the interactive dashboard. For each request, the necessary information for processing and visualizing the data was identified, providing guidance for the subsequent conceptual modeling.

### OLAP Session

**Query 1:** *What are the top 5 sectors by number of SELL transactions in China during 2024?*

- **TransactionType** (filter on SELL)
- **Date** (filtering by year 2024)
- **Sector**
- **Country** (filter on China)

**Query 2:** *What are the top 5 industries by number of BUY transactions in Q4 of 2024?*

- **TransactionType** (filter on BUY)
- **Date** → derivation of **Quarter**
- **Industry**

**Query 3:** *Rank all quarters of 2024 by total number of transactions (BUY + SELL).*

- **TransactionType**  $\in \{\text{BUY, SELL}\}$
- **Date** → derivation of **Quarter**

### Interactive Dashboard – Time Analysis

#### Time Filter

- Configurable range based on **Date**, with default values:
  - Start date: 01/01/2024
  - End date: 31/12/2024

**Line Chart** Displays the total number of transactions (BUY + SELL) over time:

- Date
- TransactionType

**Bar Charts** Three bar charts showing the count of transactions for:

- Top 3 Symbol
- Top 5 Sector
- Top 5 Industry

From the overall analysis of the requirements, the following **relevant information** emerged, which must be represented in the conceptual model:

- **Date** of the transaction, from which to derive quarter and year;
- **TransactionType** (BUY, SELL);
- **Symbol** of the company involved, along with its **Sector** and **Industry**;
- **Country** of origin of the company.

These elements will form the foundation for the upcoming conceptual design phase. In particular, a comprehensive *fact schema* will be initially constructed, including all potentially useful dimensions. Subsequently, *pruning* and *grafting* techniques will be applied to remove or refine unnecessary components, yielding an optimized version fully aligned with the identified analytical requirements.

## 2 Conceptual Design

The conceptual design represents a core phase in the construction of a data mart. In this phase, the multidimensional model is defined, describing the information to be analyzed—organized into facts, measures, and dimensions—at a conceptual level.

In this project, the conceptual model was developed following the guidelines of the *Dimensional Fact Model (DFM)*, as suggested in the reference literature. This model allows for an explicit representation of facts (observable events), quantitative measures, and analysis dimensions, including possible hierarchies.

### Fact to Be Modeled

The central fact to be modeled is the **financial transaction**, understood as an elementary event corresponding to a BUY or SELL operation involving a specific stock on a specific date.



## Building an Attribute Tree

Typically, this phase starts from a source conceptual schema (such as an E-R diagram), but this information was not available. Consequently, although not perfectly rigorous, the following structure was obtained:

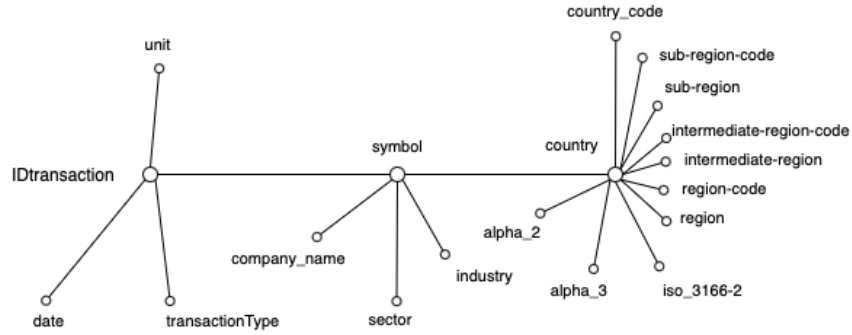


Figure 2: Attribute Tree

## Prune and Graft the Attribute Tree

The attribute tree from the previous step represents all the information logically reachable from the **Transaction** fact. However, not all components of the tree are necessary to meet the identified information requirements. Therefore, a process of *pruning* and *grafting* is applied to simplify the conceptual structure, keeping only the elements relevant for analysis.

**Pruning** (tree trimming): Nodes not involved in OLAP queries or required by the dashboard were removed. The **company\_name** was retained as it is useful for visualization purposes.

Removed attributes:

- **alpha-2, alpha-3, country-code, iso\_3166-2** (unused ISO codes)
- **region, sub-region, intermediate-region** and their corresponding codes (not used in the analysis)

**Grafting** (tree restructuring): The attribute **country** was moved from the **Company** dimension directly to the **Transaction** fact to make it immediately available for geographical analysis. This grafting reduces the tree's depth and enables a more straightforward modeling of dimensions, aligned with the analytical needs of the project.

The result is a simplified subset of the attribute tree, which serves as the foundation for building the informational dimensions in the next section. This

simplified version ensures a clearer, more efficient conceptual model aligned with the project’s goals.

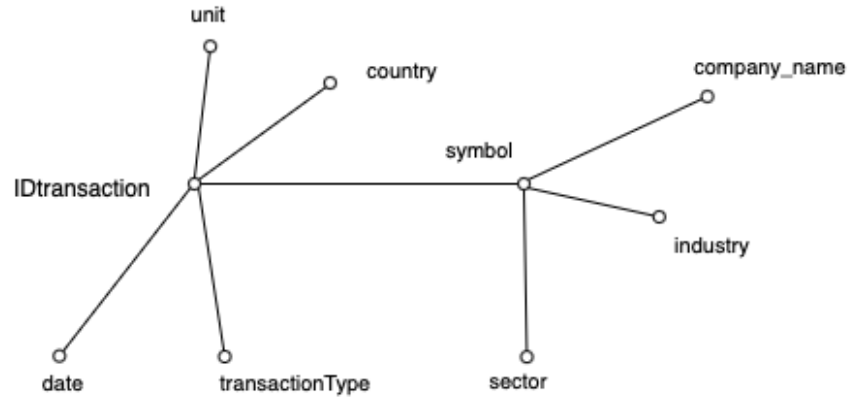


Figure 3: Attribute tree after pruning and grafting

## Definition of Dimensions and Measures

Following the pruning and grafting process of the attribute tree, it is now possible to clearly define the informational dimensions and the measures that will form the final multidimensional model.

**Informational Dimensions** The following key dimensions were identified:

- **Date:** derived from the `date` attribute in the `Transaction` fact. Since OLAP queries only reference `quarter` and `year`, the time dimension has been simplified to include only these two levels, omitting day and month.
- **TransactionType:** represents the type of operation (BUY, SELL), as indicated by the corresponding attribute.
- **Symbol:** represents the traded stock and includes, in addition to the identifier, the `sector` and `industry` attributes needed for dashboard aggregations.
- **Country:** represents the country of origin of the company and is used for geographical analysis of the transactions.

**Granularity** The granularity of the fact is no longer at the individual transaction level (i.e., each transaction identified by a unique ID), but **at a more aggregated level**.

In the final model, the level of detail corresponds to transactions performed:

by a given company (**Symbol**),  
 located in a specific country (**Country**),  
 during a given quarter (**Date**),  
 and of a certain type (**TransactionType**).

This granularity allows for a more efficient modeling aligned with the analytical needs defined in the queries.

**Measures** The main measure is the number of transactions, represented as:

- **numberOfTransactions**: obtained by counting transactions (aggregation: COUNT).

To enrich the visual analyses in the dashboard, a second measure is also included:

- **sumUnit**: total number of units traded, calculated as the sum of the **unit** attribute (aggregation: SUM).

Table 1: Measure glossary of the Transaction schema

| Measure                     | Definition            |
|-----------------------------|-----------------------|
| <b>numberOfTransactions</b> | COUNT(*)              |
| <b>sumUnit</b>              | SUM(Transaction.unit) |

These two measures, combined with the defined dimensions, provide complete coverage of the information requirements outlined in the assignment, offering flexibility for both OLAP analysis and the generation of interactive dashboard visualizations.

## Fact Schema Creation

Based on the identified dimensions and measures, it is now possible to define the *fact schema*, i.e., the logical schema that represents the central fact (**Transaction**) and its connections to the informational dimensions.

The fact is modeled through a central table that records, for each combination of dimensional attributes, the corresponding quantitative measures. In particular, the fact table contains:

- **numberOfTransactions**: aggregated transaction count
- **sumUnit**: total number of units traded

The dimensions linked to the fact include:

- **quarter**, which is linked to **year**, both derived from the **date**. The raw date is removed as it is redundant.
- **transactionType**, which distinguishes between BUY and SELL operations.
- **symbol**, which identifies the traded stock and is associated with the attributes **sector**, **industry**, and **company\_name**. Although **company\_name** is part of the original dataset, it is not modeled as a standalone dimension since it is not used in aggregations or filters. However, it remains useful for visualization purposes and is thus modeled as a **non-dimensional attribute** functionally dependent on the **Symbol** dimension.
- **country**, associated with the company conducting the transaction, modeled as a geographical dimension.

The result is a well-structured conceptual pattern in which each foreign key in the fact table refers to an analytical dimension, ensuring a clear, efficient, and OLAP-friendly model.

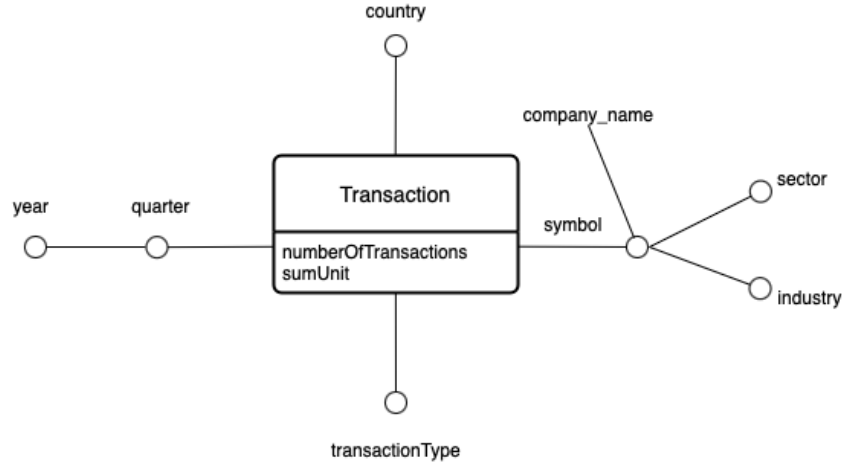


Figure 4: Fact Schema

### 3 Logical Design

The logical design of the data mart consists of defining the logical schema that implements the previously developed conceptual model. In this project, a *ROLAP* (Relational OLAP) approach was adopted, based on the *star schema*, in which a central fact is connected to multiple denormalized dimensions.

The *star schema* ensures query simplicity, good performance for analytical queries, and an intuitive structure for building dashboards and reports. The fact table contains the quantitative measures and foreign keys referencing each dimension.

## Star Schema

Figure 5 shows the *star schema* resulting from the logical design.

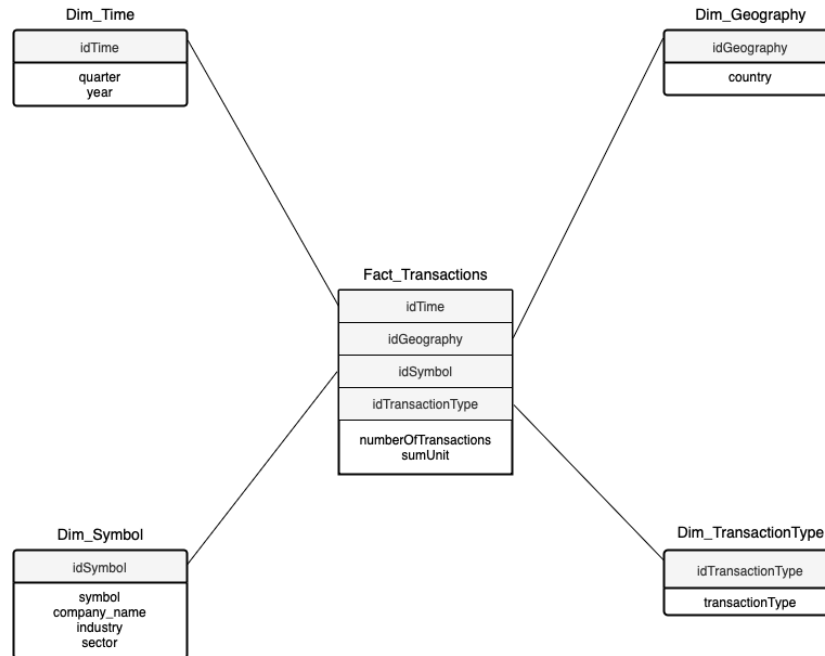


Figure 5: Star Schema

## Table Descriptions

**Fact\_Transaction** Represents the central fact, i.e., the aggregation of financial transactions. It includes two main measures:

- `numberOfTransactions` (COUNT)
- `sumUnit` (SUM of traded units)

Dimensions are connected through foreign keys:

- `idTime`
- `idGeography`
- `idSymbol`
- `idTransactionType`

**Dim.Date** Contains the time-related information necessary for analysis. In this project, granularity has been reduced to quarter and year levels:

- `Quarter`
- `Year`

**Dim.Symbol** Represents the company-related information associated with the traded stock. It includes attributes relevant for sectoral analysis:

- `company_name` (non-dimensional descriptive attribute)
- `industry`
- `sector`

**Dim.TransactionType** Contains the transaction type:

- `transactionType`  $\in \{\text{BUY, SELL}\}$

**Dim.Geography** Represents the geographical dimension of the country associated with the transaction:

- `country`

This logical structure, supported by a proper ETL phase, enables efficient execution of the required OLAP queries and the construction of a flexible and navigable interactive dashboard.

## 4 Implementation using Python

Unlike traditional systems, in this project no physical design has been applied. Instead, the focus was on a functional implementation using **Python**, with the goal of transforming raw CSV data into a structure that faithfully reflects the previously defined logical model.

The entire ETL (Extract, Transform, Load) process was implemented in a single Python script, which carries out the following main tasks:

- **Extraction:** data is loaded from three CSV files:
    - `account-statement-1-1-2024-12-31-2024.csv`: financial transactions
    - `symbols.csv`: company-related metadata
    - `country.csv`: not used in the ETL, but relevant for potential geographic enrichment
  - **Transformation:** raw data is cleaned, filtered, and enriched with derived fields such as `quarter`, `year`, and surrogate keys for each dimension. Four dimension tables are created:
    - `dim_time`
    - `dim_symbol`
    - `dim_transaction_type`
    - `dim_geography`
- These tables reflect the star schema described in the logical design, with each one being assigned an artificial primary key for joining purposes.
- **Loading:** the central `fact_transaction` table is constructed by aggregating transactional data across all relevant keys. Specifically, transactions are grouped by `symbol`, `transactionType`, `country`, and `quarter`, and two key measures are calculated:
    - `numberOfTransactions` (COUNT)
    - `sumUnit` (SUM)

Finally, all dimension and fact tables are exported as CSV files into a dedicated directory (`./data/star_schema/`), making them ready for OLAP analysis and dashboard integration.

This Python-based implementation ensures a clear and reproducible pipeline, suitable for educational purposes. It demonstrates how a requirement-driven data mart can be constructed from scratch using common data manipulation libraries such as `pandas`, without relying on complex ETL tools or data warehouse platforms, obtaining the following result:

|  | # idTime |   | quarter |  | # year |
|--|----------|---|---------|--|--------|
|  | 0        | 1 | Q1      |  | 2024   |
|  | 1        | 2 | Q2      |  | 2024   |
|  | 2        | 3 | Q3      |  | 2024   |
|  | 3        | 4 | Q4      |  | 2024   |

Figure 6: Dim.Time

| # idGeography |  | country |           |
|---------------|--|---------|-----------|
| 0             |  | 1       | Australia |
| 1             |  | 2       | Bahamas   |
| 2             |  | 3       | Bermuda   |
| 3             |  | 4       | Brazil    |
| 4             |  | 5       | Canada    |

Figure 7: Dim\_Geography

| # idSymbol | symbol | company_name                  | industry                      | sector             |
|------------|--------|-------------------------------|-------------------------------|--------------------|
| 0          | 1 AAL  | American Airlines Group, Inc. | Airlines                      | Industrials        |
| 1          | 2 AAMI | Acadian Asset Management Inc. | Asset Management              | Financial Services |
| 2          | 3 AAOI | Applied Optoelectronics, Inc. | Communication Equipment       | Technology         |
| 3          | 4 AAON | AAON, Inc.                    | Building Products & Equipment | Industrials        |
| 4          | 5 AAP  | Advance Auto Parts Inc.       | Auto Parts                    | Consumer Cyclical  |

Figure 8: Dim\_symbol

| # idTransactionType |  | TransactionType |          |
|---------------------|--|-----------------|----------|
| 0                   |  | 1               | BUY      |
| 1                   |  | 2               | DIVIDEND |
| 2                   |  | 3               | SELL     |

Figure 9: Dim\_TransactionType

| # idTime | # idSymbol | # idTransactionType | # idGeography | # numberOfTransactions | # sumUnit |        |
|----------|------------|---------------------|---------------|------------------------|-----------|--------|
| 0        | 1          | 6.0                 | 1             | 41.0                   | 9         | 507.0  |
| 1        | 1          | 6.0                 | 3             | 41.0                   | 10        | 440.0  |
| 2        | 1          | 26.0                | 1             | 3.0                    | 4         | 2000.0 |
| 3        | 1          | 37.0                | 1             | 20.0                   | 7         | 172.0  |
| 4        | 1          | 97.0                | 1             | 41.0                   | 1         | 115.0  |

Figure 10: Fact\_Transactions